Investigating with commands

NUnix · L3 MIAGE · CM2 · 90 minutes

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Investigating with commands

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A computer is slow. Would adding RAM help?
Check which resource is under pressure first.

We will use commands to inspect the system, find useful information, and turn it into a summary we can save.

CM2 outline

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1. Processes and resources
2. Finding files and text
3. Inputs, outputs and redirections
4. Combining commands with pipes

Observe the system → find information → redirect streams → combine commands.

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1. Processes and resources

Which programs are running, and where are the resources going?

CPU, memory and disk space

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Resource	Question	First observation
CPU	Is computation keeping a processor busy?	`top`, `ps`
RAM	Is memory pressure affecting the system?	`free -h`
Disk	Can the filesystem store more data?	`df -h .`

Adding RAM does not create free space on a full disk.

Running in the foreground

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A **process** is a running instance of a program.

```
sleep 10
```

The shell waits for this foreground command to finish before showing a new prompt.

Ctrl+C requests an interrupt of the foreground job.

Inspecting processes

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```
ps
ps -ef
```

Field	Meaning
PID	Process identifier
TTY	Associated terminal, if any
TIME	Accumulated CPU time
CMD	Command

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Checking memory usage

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```
free -h
ps -eo pid,comm,%cpu,%mem --sort=-%mem
```

- Read **available** memory; cache use is normal.
- Identify the application's process and follow it over time.
- Check logs before attributing every delay to memory.

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Checking disk usage

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Tool	Scope	Question
`df`	A filesystem	How much space is available?
`du`	A directory tree	Which directories use space?
`find`	Individual entries	Which files fit my criteria?

Start broad, then inspect a **specific file**.

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Find which directories use space

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```
df -h "$HOME"
du -h --max-depth=1 "$HOME"
```

`df` shows usage of the filesystem containing your home.
`du` shows how much space your directories use.

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From resources to files

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1. Processes and resources
2. **Finding files and text**
3. Inputs, outputs and redirections
4. Combining commands with pipes

We can inspect CPU, memory and disk usage. Now we need to locate files and find the relevant lines in their contents.

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2. Finding files and text

Where is the file, and which lines contain the information we need?

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Finding files

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```
find . -type f -name '*.txt'
```

- `.` is the **starting directory**.
- `find` also visits its subdirectories.
- `-type f` selects regular files.
- `-name '*.txt'` tests each entry's name.

The quotes let **find** receive the pattern unchanged.

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Search criteria

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Criterion	Matches
`-iname '*.txt'`	Name, ignoring case
`-user alice`	Entries owned by Alice
`-type d` / `-type l`	Directories / symbolic links
`-size +100M`	Size greater than 100 MiB
`-mtime -2`	Modified less than 48 hours ago

```
find . -type f -size +100M
```

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Checkpoint: find

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1. Find regular `.c` files below the current directory.
2. Why quote `'*.c'`?

1. `find . -type f -name '*.c'`
2. So the shell passes the pattern to `find` without expanding it first.

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A small log file

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We have located a log file. Next, we will search its contents with `grep`. Save these four lines as `app.log`:

```
10:00 INFO STARTED
10:01 ERROR DISK_FULL
10:02 ERROR TIMEOUT
10:03 ERROR DISK_FULL
```

Field	Meaning
1	Time
2	Severity
3	Event code

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Finding text with grep

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```
grep ' ERROR ' app.log
grep -r ' ERROR ' .
grep -n 'DISK_FULL' app.log
```

- Print **matching lines** from a file.
- `-r` searches directories recursively.
- `-n` includes line numbers.

Exit status: **0** = match, **1** = no match, **2** = error in GNU `grep`.

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Where the name grep comes from

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The editor ed had the command pattern:

```
g/re/p
```

- **g**: apply globally to matching lines.
- **re**: a regular expression.
- **p**: print those lines.

`grep` makes that text search available as a separate command.

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Patterns in grep

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Regular expression	Meaning
`.`	Any one character
`m*`	Zero or more repetitions of `m`
`[1-3]`	A digit from 1 to 3

```
grep 'm.*t' names.txt
grep 'toto[1-3]' names.txt
```

In a regular expression, * repeats the **preceding item**.

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Who interprets this pattern?

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```
printf '%s\n' *.log
grep 'DISK.*' app.log
find . -name '*.log'
```

1. The **shell** expands a filename pattern.
2. **grep** interprets a regular expression.
3. **find** interprets a quoted filename pattern.

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From matching lines to saved results

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1. Processes and resources
2. Finding files and text
3. **Inputs, outputs and redirections**
4. Combining commands with pipes

We can find files and select useful lines. How can a command read from a file and save its output or errors?

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3. Inputs, outputs and redirections

Where does a command read, where does it write, and did it succeed?

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Input, output and status

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Channel	Number	Usual terminal behaviour
`stdin`	**0**	Read keyboard input
`stdout`	**1**	Display normal output
`stderr`	**2**	Display diagnostic messages

The **exit status** is a number reported when a command finishes. It is separate from the text it prints.

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Exit status

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```
true
echo "$?"
false
echo "$?"
```

0 means success / true. **Nonzero** means failure / false.
$? is the status of the **most recent command**.

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Reading input from a file

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```
sort < names.txt
```

The shell opens `names.txt` as **standard input** for `sort`.

`sort` writes the sorted lines to standard output, which still points to the terminal.

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Saving output

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```
sort names.txt > sorted.txt
echo 'Zoé' >> sorted.txt
```

Operator	Effect on the destination file
>	Create it, or **replace** its contents
>>	Create it, or **append** to its contents

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Redirecting errors

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```
find . -name '*.txt' > matches.txt 2> errors.txt
```

- > redirects **stdout**, descriptor 1.
- 2> redirects **stderr**, descriptor 2.
- Normal results and diagnostics go to separate files.

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Redirections are applied from left to right

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```
ls notes.txt missing > all.txt 2>&1
ls notes.txt missing 2>&1 > out.txt
```

Command	stdout ends up…	stderr ends up…
First	In `all.txt`	In `all.txt`
Second	In `out.txt`	At the terminal

2>&1 copies stdout's destination **at that moment**.
It does not make stderr follow later changes to stdout.

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Why did the file become empty?

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```
sort names.txt > names.txt
```

The shell opens and truncates `names.txt` **before** starting `sort`. The input is already gone.

Write to a **different file**, then inspect the result.

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From one command to a pipeline

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1. Processes and resources
2. Finding files and text
3. Inputs, outputs and redirections
4. **Combining commands with pipes**

We can choose the input and output of one command. Now we will connect commands to count and rank the errors in our log.

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4. Combining commands with pipes

How can one command use another command’s output?

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A pipe between two commands

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```
grep ' ERROR ' app.log | wc -l
```

`grep` writes matching lines to its standard output.
The pipe connects that output to wc's standard input.
The result is the **number of matching lines**.

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Connecting programs like garden hoses

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Doug McIlroy described connecting programs so that one program’s output could become another’s input.

```
grep ' ERROR ' app.log | wc -l
```

Each tool has a small job.
The shared text stream lets us combine them.

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Pipeline step 1: extract error codes

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```
grep ' ERROR ' app.log |
    cut -d ' ' -f 3
```

```
DISK_FULL
TIMEOUT
DISK_FULL
```

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Pipeline step 2: count each code

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```
grep ' ERROR ' app.log |
    cut -d ' ' -f 3 |
    sort | uniq -c | sort -nr
```

```
      2 DISK_FULL
      1 TIMEOUT
```

`uniq -c` counts **adjacent** equal lines. Sort first.

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Save and display output with tee

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```
sort names.txt | tee sorted.txt
```

`tee` copies its input to a file **and** standard output.

Think of a T-shaped pipe fitting: the stream has two outlets.

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Combining arguments and redirections

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```
sort < names.txt > sorted.txt
sort < names.txt | head -n 3 > first-three.txt
```

- `< names.txt` selects the input stream.
- `-n 3` remains an **argument** to head.
- `> first-three.txt` selects the output file.

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Checkpoint: redirections

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1. Which operator appends to a file?
2. What does a `|` b connect?
3. Does a `> out.txt` also redirect standard error?

1. >>
2. a's stdout to b's stdin.
3. No. Use 2> to redirect stderr.

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The exit status of a pipeline

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```
false | sort
echo "$?"
set -o pipefail
false | sort
echo "$?"
```

Output: **0**, then **1**.
Bash normally reports the last pipeline command's status.

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CM2 summary: commands and pipelines

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1. **Processes and resources:** inspect CPU, memory and disk usage.
2. **Finding files and text:** locate files with `find`, select lines with `grep`.
3. **Inputs, outputs and redirections:** read files, save results, separate errors.
4. **Combining commands with pipes:** extract, count and rank log entries.

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Optional reference

More search expressions and interactive process control

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Combining criteria

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```
find /usr -name '*.c'
find . -user alice -name '*.c'
find . -not \( -name '*.obj' -o -name '*.map' \)
```

- Adjacent tests use **AND**. `-a` makes it explicit.
- `-o` means **OR**. `-not` negates a test.
- Escaped parentheses group tests for `find`.

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Running a command on matches

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```
find . -type f -name '*.txt' -exec wc -l {} \;
```

- `-exec` starts the command to run.
- `{}` stands for the current matching path.
- `\;` ends the command without ending the shell's command line.

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Running in the background

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```
sleep 60 &
jobs
fg %1
```

- `&` lets the shell return to the prompt immediately.
- `jobs` lists jobs managed by this shell.
- `fg %1` brings job **1** to the foreground.

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Suspending and resuming a job

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Action	Effect
Ctrl+Z	Suspend the foreground job
`bg %1`	Resume job 1 in the background
`fg %1`	Resume job 1 in the foreground

A **job number** belongs to a shell.
A **PID** identifies a process in the system.

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Sequential and asynchronous execution

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```
who | sort; date
sleep 10 &
date
```

- `;` runs the next command after the previous one finishes.
- `&` starts work asynchronously.
- The final `date` can run while `sleep` is still active.

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Signals and end of input

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Action	Meaning
`kill PID`	Send `SIGTERM` to a process
Ctrl+C	Send `SIGINT` to the foreground process group
`Ctrl+\`	Send `SIGQUIT` to the foreground process group
Ctrl+Z	Send `SIGTSTP` to suspend the foreground group
Ctrl+D	Terminal end-of-input action

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